

Multimedia Techniques & Applications 2025

U2315

National Taipei University



Final Exam

1. Answer the following questions about color science:

- (a) What is the spectral power distribution (SPD) of a light? (4%)
SPD is a description of how the intensity of light varies with its wavelength.
- (b) What is the major disadvantage of using SPD to represent color in digital multimedia? (4%)
SPDs are too cumbersome for representing the color.
- (c) Describe the Tristimulus Theory (4%)
All visible SPDs (colors) can be approximated with three values, giving the weights of each of the three components.

2. Provide example scenarios where the CMYK and YUV color models are preferable to the RGB model, respectively. (6%)

- **The CMYK color space is more effective to represent color ink (3%).**
- **The YUV color space is useful when we need to separate the brightness information of an image from its color (3%).**

Note: other correct answers are acceptable.

3. The following figure shows the procedure of JPEG compression. Answer the following questions:

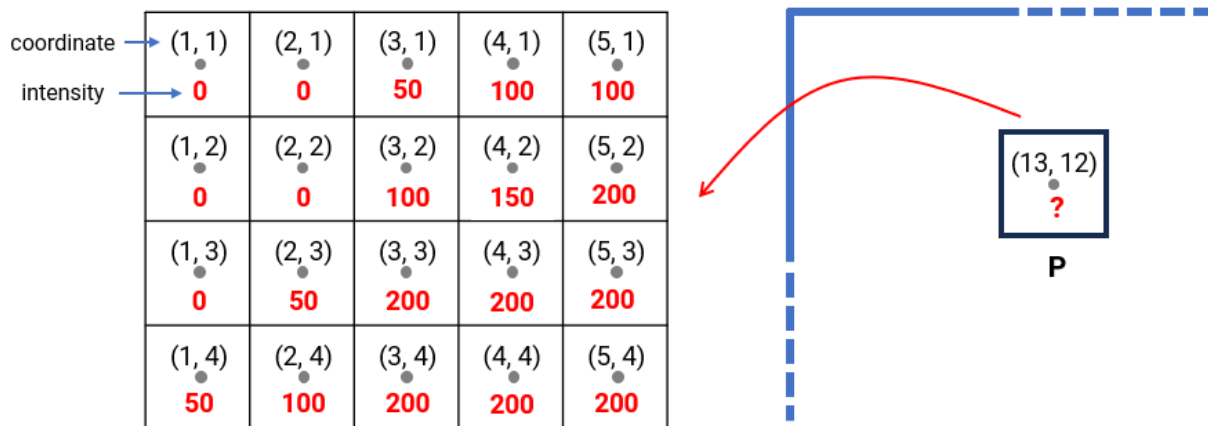


- (a) What is the purpose of converting the RGB color space to YCbCr color space? (4%)
Firstly, humans are more sensitive to brightness (Y) than to color components (Cb and Cr). Secondly, Cb and Cr contain lower-frequency information and exhibit greater spatial consistency. Therefore, we leave Y uncompressed and apply compression only to Cb and Cr. (full score can be earned by correctly addressing either point.)
- (b) What is the purpose of dividing the image into smaller blocks? (4%)
Smaller blocks in an image tend to have higher spatial coherence.
- (c) Which step makes JPEG a lossy compression? (3%)
Quantization (3%)

4. Assume you are scaling a gray-scale image from 5×4 to 25×20 using inverse mapping, as illustrated in the figure below (the coordinate represents the center of the pixel):

- (a) What will be the intensity at $P(13, 12)$ using the nearest-neighbor method? (4%)
100 (The closest pixel after scaling is (3, 2)).

- (b) What will be the intensity at $P(13, 12)$ using the bilinear interpolation? (4%)
 $0*0.4*0.6 + 100*0.6*0.6 + 50*0.4*0.4 + 200*0.6*0.4 = 92$



5. Describe why Box filters and Gaussian filters cause edge blurring. (4%)
Box and Gaussian filters rely solely on spatial distance for weighting during filtering, which causes them to smooth over edges.
6. Imagine you've taken an over-exposed (too bright) photo of a moving scene with a digital camera. Which camera settings can you modify to darken the image, and what are the potential side effects of these adjustments? (8%)
- **Use a shorter shutter speed:** this will diminish the motion blur effect, if that effect is desired.
 - **Decrease the aperture size:** this will reduce the depth of field effect, if that is the intended outcome.
7. Describe the camera limitation that the High Dynamic Range (HDR) Imaging technique is designed to address. Then, provide a brief summary of the main steps involved in the HDR algorithm. (6%)
Due to hardware limitations, cameras are unable to capture the full spectral content and dynamic range. The process involves four steps:
- **Capture images with multiple exposures** (by setting various shutter speeds)
 - **Perform image alignment algorithms** to align images with various exposures
 - **Recover the response curve of the camera** using optimization
 - **Remove ghost or flare**
8. Explain the purpose of a tone mapping algorithm. (4%)
Tone mapping algorithms designed to map scene luminances (HDR) to display luminances (LDR) in a way that aligns with human visual perception.
9. Describe how diffuse, glossy, and perfect specular surfaces reflect light. Then, provide real-world examples for each type of material. (9%)
- **Diffuse:** incoming light is scattered toward every direction upon reflection (2%).
Example: concrete walls, papers, unpolished wood, etc (1%).

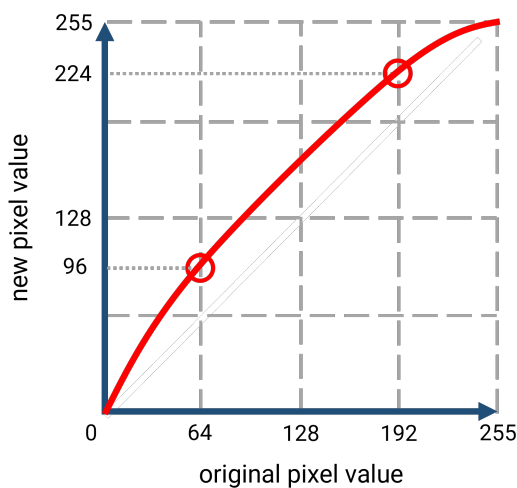
- **Glossy:** incoming light is predominantly scattered in the ideal reflection direction, where the angle of reflection matches the angle of incidence, with minor deviations (2%). Examples: polished metals or ceramics (1%).
- **Specular:** light is reflected such that the angle of reflection equals the angle of incidence (2%). Example: mirror (1%).

Note: answer by drawing graph is acceptable.

10. Explain how environment maps effectively store the directionally varying light of a scene to illuminate virtual objects. (6%)

Each pixel's two-dimensional coordinate in an environment map corresponds to a specific spherical direction, and its pixel value indicates the incoming radiance from that direction.

11. Answer the following questions about pixel point processing using the figures below:



original image
(lower contrast)

enhanced image
(higher contrast)

- (a) The left figure shows a color curve used for adjusting contrast. Explain, with justification, how the contrast is influenced — whether it increases or decreases — within the intensity intervals $[0, 64]$ and $[192, 255]$ in the original image. (4%)

The contrast within the range $[0, 64]$ is increased as it is expanded to a broader range $[0, 96]$ (2%). Conversely, the contrast within the range $[192, 255]$ is decreased due to its compression into a narrower range $[224, 255]$ (2%).

- (b) For the right figure, describe how histogram equalization can be used to enhance image contrast. (4%)

We "We begin by constructing the histogram, which represents the intensity distribution of the original image. Next, we remap this distribution to a broader range to improve contrast. Finally, we generate the enhanced image by adjusting the intensity values based on the mapping.

12. A single point on a camera sensor can capture only one value. Explain how a color filter array works to produce color images. (6%)

The sensors on a camera sensor array are organized in a special pattern where each sensor only captures one channel among R, G, and B. The missing channels are estimated (interpolated) by the neighboring sensors.

13. Polarized 3D glasses play a crucial role in viewing 3D movies. Explain how they work to create a sense of stereo vision. (6%)

Polarized 3D glasses enable stereo vision in 3D movies by ensuring that each eye sees a slightly different image, mimicking how our eyes perceive depth in the real world. This is achieved by:

- For the display: two images, one for the left eye and one for the right, are projected onto the screen simultaneously. These images are polarized differently: one image is polarized horizontally and the other vertically.
- For the glasses: the lenses of the 3D glasses are specially designed to match these polarizations. Each lens only allows light with a specific polarization to pass through. The left lens allows only the image polarized for the left eye, and the right lens allows only the image for the right eye.

14. One of the main challenges in virtual reality (VR) is the high rendering cost. Explain why VR games require significantly more rendering computations compared to traditional PC or console games. (6%)

VR games requires to render more pixels due to higher resolutions and dual views at higher frame rates to prevent motion sickness. (answering only one factor will earn a score of 4%)